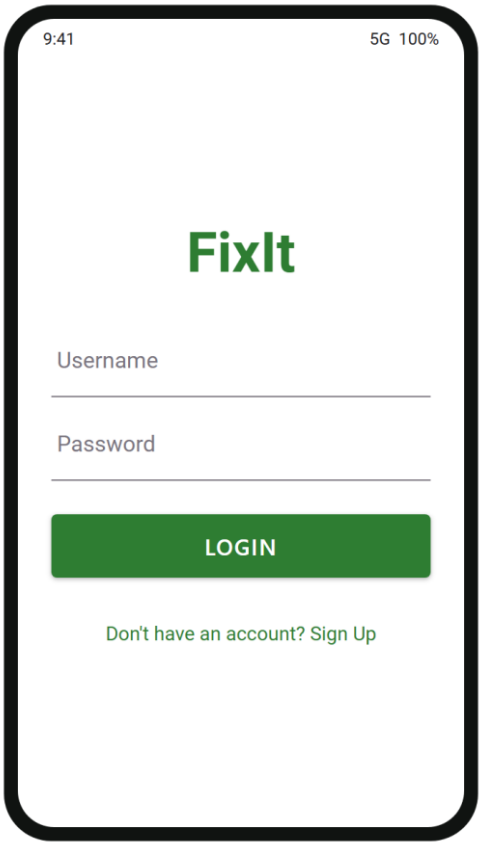
Person 1: Login

FixIt, Landlord–Tenant Maintenance App. Complete work for the Login screen.



activity_login.xml, portrait layout (preview of the screen)

What it does

- Login screen with a FixIt title, username and password fields, a Login button and a “Sign Up” link.
- Empty field: toast “Please fill in all fields”.
- Wrong credentials: toast “Username or password is incorrect”.
- Success: saves user id and role to SharedPreferences, then opens Tenant Home or Landlord Home depending on the role.
- User entity with user_id, name, username, password, email, role and unit_number. The username is unique.
- UserDao with insert, login, getByUsername (for Sign Up) and getById (for Request Detail).

Screen components

Component	ID	Notes
TextView: app title	none	“FixIt”, 40sp, bold, green
EditText: username	etUsername	single line
EditText: password	etPassword	inputType textPassword
Button: Login	btnLogin	full width
TextView: sign up link	tvGoSignup	clickable, opens SignupActivity

Test accounts (temporary)

Created automatically the first time the app opens. Remove the seedDemoUsers() code from LoginActivity once Sign Up works.

Role	Username	Password
Tenant	`tenant1`	`tenant123`
Landlord	`landlord1`	`landlord123`

For the rest of the team

- Read the logged-in user with `LoginActivity.PREFS_NAME` (“fixit_prefs”), `KEY_USER_ID` and `KEY_ROLE`.
- Roles are `User.ROLE_TENANT` and `User.ROLE_LANDLORD`.
- Passwords are stored as plain text, as the spec has them.

LoginActivity.java

Path: `app/src/main/java/com/fixit/app/LoginActivity.java`

```
package com.fixit.app;

import android.content.Context;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

public class LoginActivity extends AppCompatActivity {

    public static final String PREFS_NAME = "fixit_prefs";
    public static final String KEY_USER_ID = "user_id";
    public static final String KEY_ROLE = "role";

    private final ExecutorService executor = Executors.newSingleThreadExecutor();

    private EditText etUsername;
    private EditText etPassword;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        Button btnLogin = findViewById(R.id.btnLogin);
        TextView tvGoSignup = findViewById(R.id.tvGoSignup);

        seedDemoUsers();

        btnLogin.setOnClickListener(v -> attemptLogin());
        tvGoSignup.setOnClickListener(v ->
                startActivity(new Intent(this, SignupActivity.class)));
    }

    // TEMPORARY: demo accounts so Login can be tested before Sign Up exists. Remove before
final submission.
    private void seedDemoUsers() {
        executor.execute(() -> {
            UserDao dao = AppDatabase.getInstance(getApplicationContext()).userDao();
            if (dao.getByUsername("tenant1") == null) {
                User tenant = new User();
                tenant.name = "Demo Tenant";
                tenant.username = "tenant1";
                tenant.password = "tenant123";
                tenant.email = "tenant1@example.com";
                tenant.role = User.ROLE_TENANT;
                tenant.unitNumber = "A-101";
                dao.insert(tenant);
            }
            if (dao.getByUsername("landlord1") == null) {
                User landlord = new User();
                landlord.name = "Demo Landlord";
                landlord.username = "landlord1";
```

```
                landlord.password = "landlord123";
                landlord.email = "landlord1@example.com";
                landlord.role = User.ROLE_LANDLORD;
                dao.insert(landlord);
            }
        });
    }

    private void attemptLogin() {
        String username = etUsername.getText().toString().trim();
        String password = etPassword.getText().toString();

        if (username.isEmpty() || password.isEmpty()) {
            Toast.makeText(this, "Please fill in all fields", Toast.LENGTH_SHORT).show();
            return;
        }

        executor.execute(() -> {
            User user = AppDatabase.getInstance(getApplicationContext())
                    .userDao().login(username, password);
            runOnUiThread(() -> {
                if (user == null) {
                    Toast.makeText(this, "Username or password is incorrect",
                            Toast.LENGTH_SHORT).show();
                } else {
                    onLoginSuccess(user);
                }
            });
        });
    }

    private void onLoginSuccess(User user) {
        getSharedPreferences(PREFS_NAME, Context.MODE_PRIVATE)
                .edit()
                .putInt(KEY_USER_ID, user.userId)
                .putString(KEY_ROLE, user.role)
                .apply();

        Class<?> home = User.ROLE_LANDLORD.equals(user.role)
                ? LandlordHomeActivity.class
                : TenantHomeActivity.class;
        startActivity(new Intent(this, home));
        finish();
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        executor.shutdown();
    }
}
```

activity_login.xml

Path: `app/src/main/res/layout/activity_login.xml`

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center_vertical"
        android:orientation="vertical"
        android:padding="24dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginBottom="32dp"
            android:gravity="center"
            android:text="FixIt"
            android:textColor="@color/fixit_primary"
            android:textSize="40sp"
            android:textStyle="bold" />

        <EditText
            android:id="@+id/etUsername"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:autofillHints="username"
            android:hint="Username"
            android:imeOptions="actionNext"
            android:inputType="text"
            android:maxLines="1"
            android:minHeight="48dp" />

        <EditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="12dp"
            android:autofillHints="password"
            android:hint="Password"
            android:imeOptions="actionDone"
            android:inputType="textPassword"
            android:maxLines="1"
            android:minHeight="48dp" />

        <Button
            android:id="@+id/btnLogin"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="24dp"
            android:text="Login"
            android:textSize="16sp" />

        <TextView
            android:id="@+id/tvGoSignup"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
            android:clickable="true"
            android:focusable="true"
            android:gravity="center"
            android:minHeight="48dp"
            android:text="Don't have an account? Sign Up"
```

```
            android:textColor="@color/fixit_primary"
            android:textSize="14sp" />

    </LinearLayout>
</ScrollView>
```

User.java

Path: `app/src/main/java/com/fixit/app/User.java`

```
package com.fixit.app;

import androidx.room.ColumnInfo;
import androidx.room.Entity;
import androidx.room.Index;
import androidx.room.PrimaryKey;

@Entity(tableName = "users", indices = {@Index(value = "username", unique = true)})
public class User {

    public static final String ROLE_TENANT = "tenant";
    public static final String ROLE_LANDLORD = "landlord";

    @PrimaryKey(autoGenerate = true)
    @ColumnInfo(name = "user_id")
    public int userId;

    public String name;

    public String username;

    public String password;

    public String email;

    public String role;

    @ColumnInfo(name = "unit_number")
    public String unitNumber;
}
```

UserDao.java

Path: `app/src/main/java/com/fixit/app/UserDao.java`

```
package com.fixit.app;

import androidx.room.Dao;
import androidx.room.Insert;
import androidx.room.Query;

@Dao
public interface UserDao {

    @Insert
    long insert(User user);

    @Query("SELECT * FROM users WHERE username = :username AND password = :password LIMIT 1")
    User login(String username, String password);

    @Query("SELECT * FROM users WHERE username = :username LIMIT 1")
    User getByUsername(String username);

    @Query("SELECT * FROM users WHERE user_id = :userId LIMIT 1")
    User getById(int userId);
}
```